

RESQ-PPO: MULTI-AGENT PROXIMAL POLICY OPTIMIZATION FOR CRISIS ROUTING

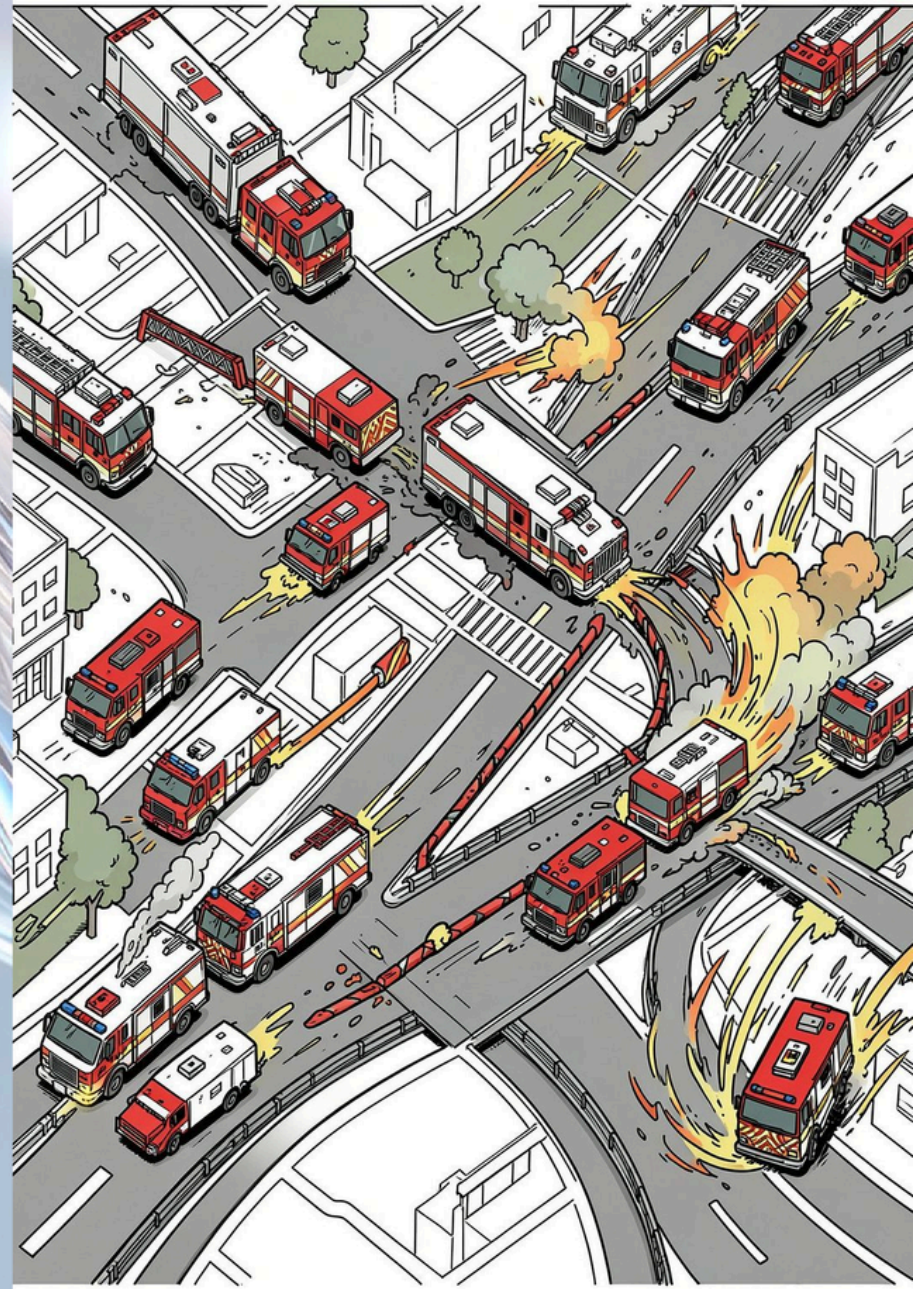
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Introduction

- Crisisresponse needsfast, adaptive routing.
- Multiple agents (ambulance, fire, police) must coordinate.
- Traffic, closures, evolving hazards make planning hard.
- RESQ-PPO uses multi-agent RL to learn routing strategies.

Problem Statement

- Shortest-pathalgorithmsdon't adapt wellto dynamic crises.
- Multiple agents can conflict or create bottlenecks.
- Need an intelligent, dynamic routing system for emergencies.

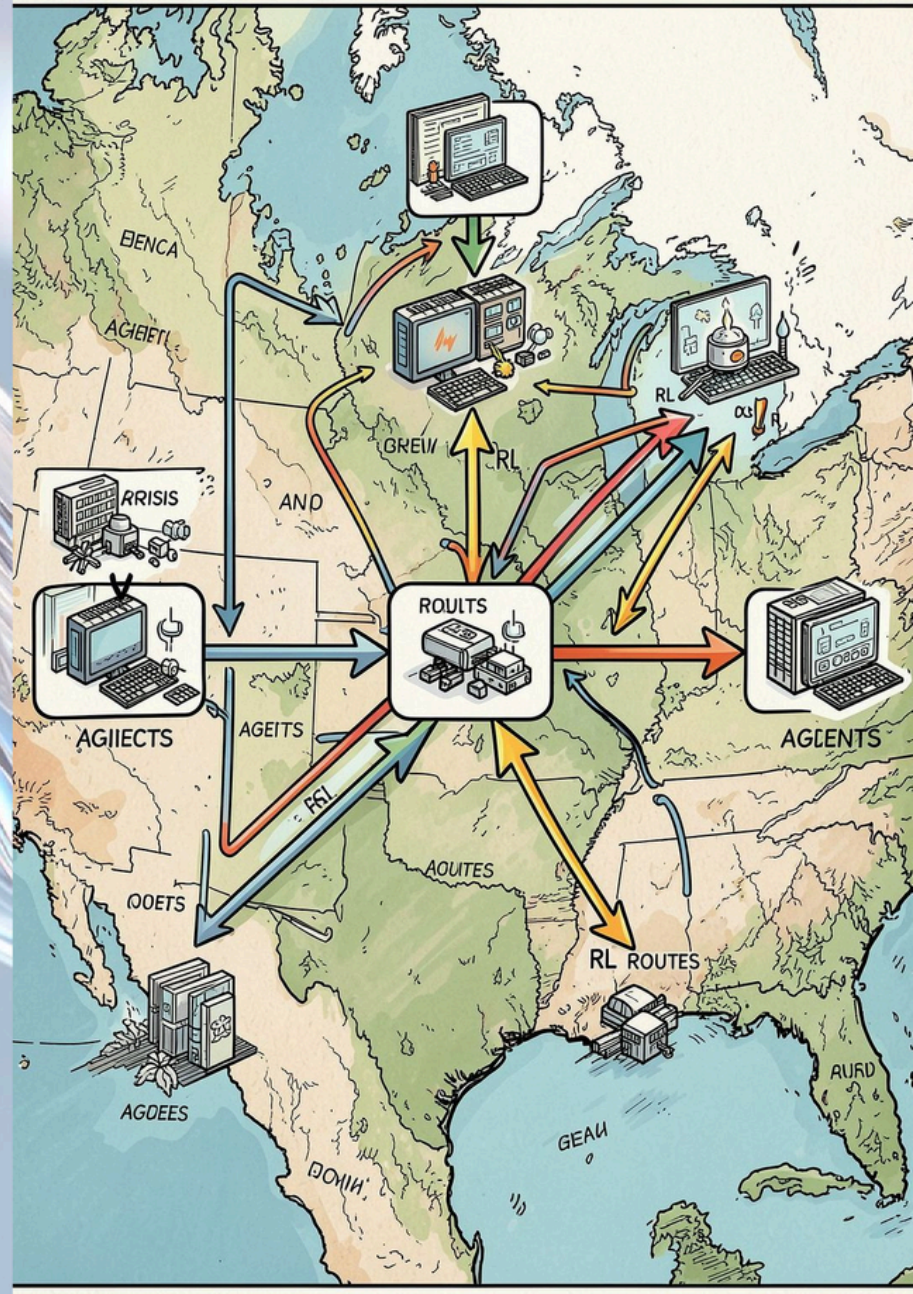


Objectives

- Intelligent crisis routing
- Coordinate multiple agents
- Dynamic routing optimization
- Reduce travel time/conflicts
- Interactive web visualization

Proposed Solution

Crisis Environment→Multiple Agents → Environment State → Multi-Agent PPO → Action Selection → Optimized Routes → Web Visualization



System Architecture



Architecture maps report components exactly. (Components not in report: [none introduced])

Dataset

Dataset name: [RESQ-PPO Synthetic Crisis Routing Dataset]

Source: [Simulated data generated for the RESQ-PPO project]

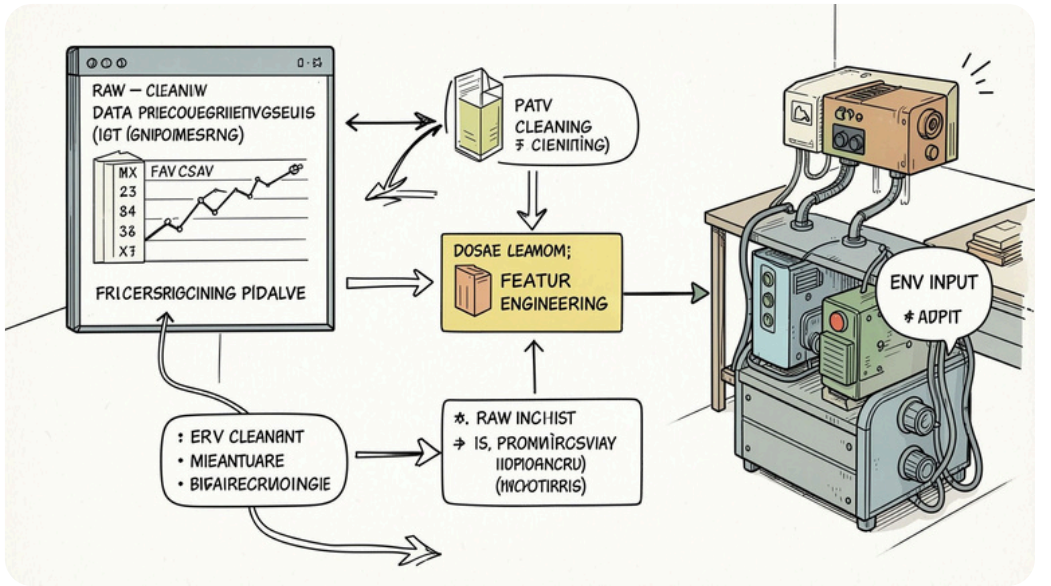
Number of records: [1,000 simulated records]

Number of features: [10]

Important columns: [agent_id, source, destination, traffic_level, road_condition, distance, crisis_level, latitude, longitude, route_cost]

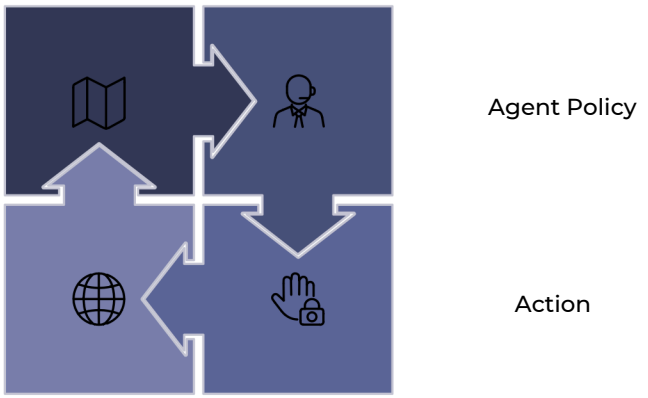
Type: [Synthetic / Simulated Data]

Usage: Environment inputs for state, traffic/road attributes, crisis markers



Raw Data → Data Cleaning → Preprocessing → Environment Input → RL Model

Multi-Agent PPO (overview)



01
What is PPO?

Policy-based RL with constrained updates for stability.

02
Multi-Agent PPO

Multiple agents learn concurrently; coordinate via shared or individual policies.

03
Training Flow

State → Agent Policy → Action → Environment → Reward → Policy Update



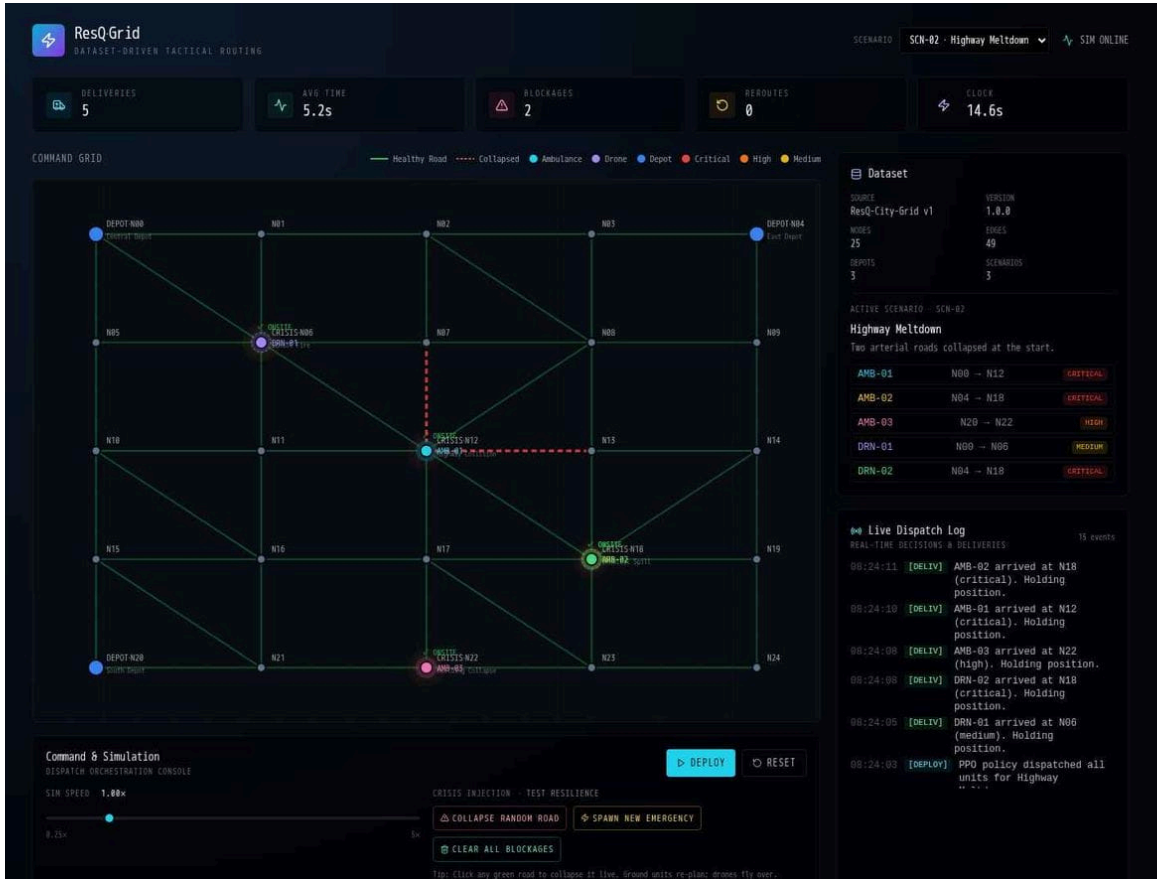
Website — RESQ-PPO Interface

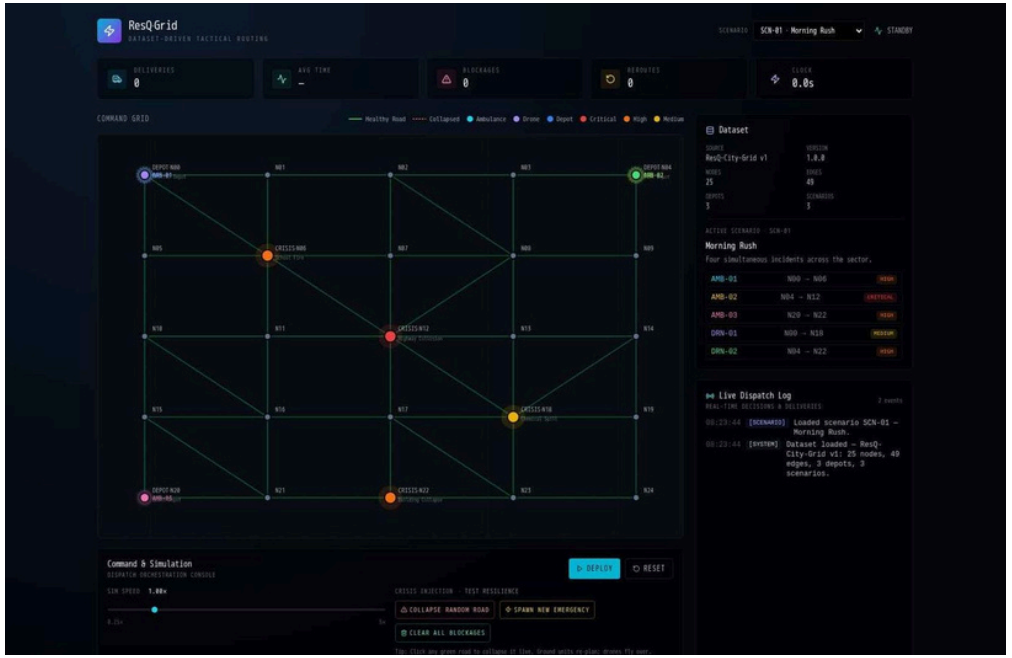
Actualprojectwebsitescreenshot— useprovidedimages

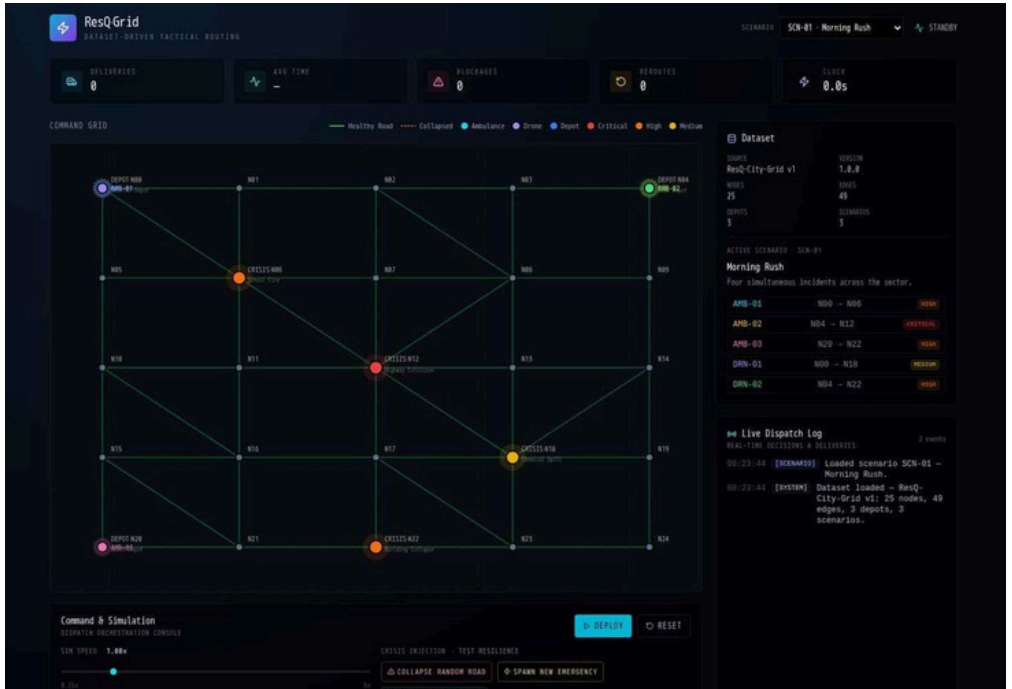
Callouts: Map/Environment • Agents • Routes • Crisis locations • Controls • Results

Results & Output

Actual screenshots and metrics from the project (use report values)







Source Code & Workflow

```
# Environment creation (snippet)
env = RESQEnvironment(data=env_data)
env.reset()
# PPO agent snippet
agent = MultiAgentPPO(policy=PolicyNet, lr=3e-4)
# Training loop (short)
forep in range(episodes):
  states = env.get_states()
  actions = agent.select_actions(states)
  next_states, rewards = env.step(actions)
  agent.update(states, actions, rewards, next_states)
```

Workflow

Crisis Input → Environment → Multiple Agents → PPO Policy → Actions → Reward → Optimized Routing → Website Output

Integration

Trained policy exported → Web backend loads policy → Visualization updates in real time

Advantages • Future Scope • Conclusion



Advantages

- Dynamiccrisis routing
- Multi-agent coordination
- Adaptive, AI-driven decisions
- Interactive visualization



Future Scope

- [Usefuture work from report — e.g., real-time traffic, more agents, real-world trials]
- [TO BE ADDED if report lists specifics]



Conclusion

RESQ-PPOapplies Multi-Agent PPO to crisis routing. Agents learn coordinated routing in dynamic environments and results are visualized via a web interface.



Thank You